Reduced-Pauli entanglement checks for DLA-sector leakage mechanisms on IBM Heron hardware

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Abstract

Parity-sector leakage asymmetry in small Kuramoto–XY circuits is a useful hardware observable only if the mechanism can be separated from layout, prepared-state, and readout artefacts. We report a cost-bounded reduced-Pauli entanglement/tomography check for promoted Phase 3 DLA-parity and Fisher-information-modified Kuramoto–XY circuit families. The approved `ibm_marrakesh` execution used physical qubits $[1, 2, 3, 4]$, 162 main circuits, four readout calibration circuits, 2048 shots per main circuit, and 8192 shots per readout circuit. The submitted IBM jobs were `d86g7h1789is738vkreg` and `d86ggpis46sc73f6v170`. A preregistered reducer produced 54 observable rows. The mean absolute measured-minus-exact deviation is 0.1299 and the maximum absolute deviation is 0.5561. The largest deviations are concentrated in transverse two-qubit correlators, especially DLA odd-sector `XX`/`YY` edge channels, while the FIM $\lambda_{\mathrm{fim}} = 4$ circuit deviates more strongly than the $\lambda_{\mathrm{fim}} = 0$ reference. The result is a mechanism-boundary measurement: it shows structured reduced-Pauli distortions accompanying the small-system DLA/FIM hardware programme, but does not claim scalable tomography, quantum advantage, full-state reconstruction, or backend-general entanglement dynamics.

I. INTRODUCTION

Kuramoto synchronisation and XY spin Hamiltonians provide a compact route from phase-oscillator structure to gate-model quantum dynamics [1, 2]. In the present programme, the relevant small-system hardware question is not whether a broad advantage has been demonstrated. It is narrower: when sector-resolved leakage and retention observables deviate from exact Kuramoto–XY references on IBM hardware, do reduced-Pauli correlators show an accompanying structure that can constrain the mechanism?

Previous DLA-parity and FIM runs established bounded hardware observations, including backend, layout, state-preparation, and readout sensitivity. A scalar leakage metric alone cannot distinguish coherent correlator deformation from population readout effects or generic depth-dependent decoherence. This paper therefore measures a small set of preregistered reduced-Pauli observables for the same class of promoted circuits. Full tomography is

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Label	Family	Initial state	Depth
`dla_even_shallow`	DLA parity	`0011`	6
`dla_odd_shallow`	DLA parity	`0001`	6
`dla_even_signal`	DLA parity	`0011`	10
`dla_odd_signal`	DLA parity	`0001`	10
`fim_lambda0_reference`	FIM pair	`0011`	4
`fim_lambda4_feedback`	FIM pair	`0011`	4

TABLE I. Promoted source circuits for the reduced-Pauli check.

unnecessary for this question: the aim is to compare selected two-qubit correlators against exact classical references under one backend, layout, shot budget, and calibration window.

II. PROTOCOL

The source circuits are listed in Table I. Four DLA-parity circuits compare even and odd initial states at shallow and signal depths. Two FIM-pair circuits compare $\lambda_{\mathrm{fim}} = 0$ and $\lambda_{\mathrm{fim}} = 4$ at fixed initial state and depth.

For each source circuit we measure nine reduced-Pauli basis settings,

$$\{\mathtt{IIXX}, \mathtt{IIYY}, \mathtt{IIZZ}, \mathtt{IXXI}, \mathtt{IYYI}, \mathtt{IZZI}, \mathtt{XXII}, \mathtt{YYII}, \mathtt{ZZII}\}. \tag{1}$$

Each source/basis pair is repeated three times at 2048 shots, giving 162 main circuits. Four computational-basis readout states, `0000`, `0001`, `0011`, and `1111`, are measured at 8192 shots. The complete hardware block therefore contains 166 circuits. The selected physical qubits are $[1, 2, 3, 4]$ on `ibm_marrakesh`; the maximum transpiled depth in the live preflight is 388, and the maximum measurement-basis depth expansion is 1.0718.

III. ANALYSIS

For each measured basis-rotated circuit, the reducer estimates the Pauli expectation

$$\langle P \rangle = \frac{1}{N} \sum_b n_b \prod_{i \in \mathrm{supp}(P)} (-1)^{b_i}, \tag{2}$$

Circuit label	Mean signed	Mean absolute	Max absolute
`dla_even_shallow`	-0.120474	0.148396	0.314180
`dla_even_signal`	0.061095	0.063458	0.160443
`dla_odd_shallow`	0.098416	0.219708	0.515473
`dla_odd_signal`	0.153497	0.176026	0.556091
`fim_lambda0_reference`	0.030496	0.052205	0.088148
`fim_lambda4_feedback`	0.096107	0.119566	0.274155

TABLE II. Label-level measured-minus-exact deviation aggregates. Each row summarises nine reduced-Pauli observables.

where P is the reduced-Pauli label, n_b is the observed count for bitstring b, and N is the total shot count. Repetitions are grouped by source label and basis setting, then compared with exact reference expectations from the committed reference table `data/phase3_entanglement_tomography/entanglement_observable_rows_2026-05-07.csv`.

The primary output rows are committed in `data/phase3_entanglement_tomography/entanglement_tomography_rows_2026-05-20.csv`. The rows have SHA256 digest `3d18308d60fe32827bae7517f18fd71690240b105779287408c4749cb0e7dc72`.

IV. RESULTS

The approved IBM execution completed on `ibm_marrakesh`. The submitted jobs were `d86g7h1789is738vkreg` and `d86ggpis46sc73f6v170`. The analysis produced 54 observable rows with mean absolute deviation 0.1299 and maximum absolute deviation 0.5561 relative to exact references.

Table II gives the label-level aggregates. DLA circuits have larger mean absolute deviation than the FIM pair overall. Within the DLA family, the odd shallow and odd signal circuits produce the largest aggregate departures. Within the FIM pair, $\lambda_{\text{fim}} = 4$ has more than twice the mean absolute deviation of $\lambda_{\text{fim}} = 0$.

Figure 1 shows the signed measured-minus-exact deviations by source label and basis setting. The strongest pattern is not a uniform decay of all correlators. Instead, deviations concentrate in transverse two-qubit channels. The odd-sector DLA signal circuit has large

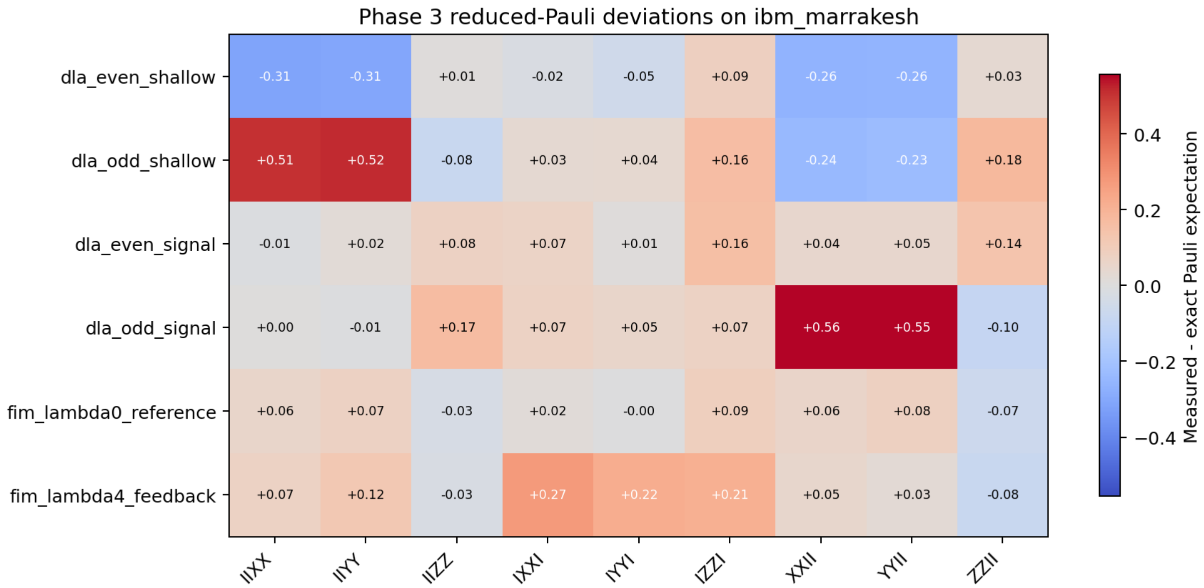

FIG. 1. Measured-minus-exact reduced-Pauli deviations for the completed `ibm_marrakesh` run. The strongest deviations concentrate in transverse edge correlators rather than in all observables uniformly.

positive shifts in `XXII` and `YYII`; the odd shallow circuit has analogous positive shifts in `IIXX` and `IIYY`. The even shallow circuit moves in the opposite direction on corresponding transverse channels.

The largest individual deviations are shown in Table III. The top four are DLA odd-sector transverse correlators. The strongest FIM-channel departure is the `IXXI` observable for the $\lambda_{\text{fim}} = 4$ circuit.

V. DISCUSSION

The main result is not that the hardware reproduces the exact reduced-Pauli reference structure. It does not. The contribution is that the departures from the exact reference are large, structured, and concentrated in specific correlator channels. This turns the earlier leakage/retention observations into a sharper hardware-mechanism question: the same circuit families that show sector-sensitive survival behaviour also show measurable correlator-level distortions.

The DLA circuits dominate the strongest deviations. The sign structure is particularly

Circuit label	Basis	Measured	Exact	Deviation
`dla_odd_signal`	`XXII`	0.431641	-0.124450	0.556091
`dla_odd_signal`	`YYII`	0.429688	-0.124450	0.554138
`dla_odd_shallow`	`IIYY`	0.488932	-0.026541	0.515473
`dla_odd_shallow`	`IIXX`	0.478841	-0.026541	0.505382
`dla_even_shallow`	`IIXX`	-0.456055	-0.141874	-0.314180
`dla_even_shallow`	`IIYY`	-0.447591	-0.141874	-0.305717
`fim_lambda4_feedback`	`IXXI`	0.175781	-0.098373	0.274155

TABLE III. Largest measured-minus-exact deviations in the reduced-Pauli rows.

important: odd-sector circuits show large positive transverse edge-correlator shifts, whereas the even shallow circuit shows large negative shifts on corresponding channels. This is not consistent with a single scalar decoherence factor multiplying every observable equally. It is more naturally read as a sector- and edge-dependent deformation under the selected backend, layout, circuit family, and calibration window.

The FIM result is more muted but still useful. The $\lambda_{\text{fim}} = 4$ feedback circuit deviates more strongly than the $\lambda_{\text{fim}} = 0$ reference. This is consistent with the earlier negative fixed-FIM hardware result: the tested digital FIM modification does not simply protect the target structure and introduces a measurable reduced-Pauli distortion channel. The result motivates adaptive FIM follow-up only under a conservative claim boundary; it does not rescue a fixed-λ_{fim} protection claim.

The readout boundary remains central. Four readout calibration states are included in the execution block, but the manuscript-level conclusion is based on unmitigated measured correlators compared with exact references. Because the largest deviations occur in transverse basis-rotated measurements, not only in computational-basis `ZZ` channels, the result is unlikely to be a pure population-readout story. It may still include basis-rotation error, layout-dependent coherent error, calibration drift, and depth-dependent decoherence. The safe interpretation is therefore mechanism-boundary evidence, not full causal attribution.

VI. CLAIM BOUNDARY

The supported claims are intentionally narrow. The run measured preregistered reduced-Pauli correlators for small Kuramoto–XY DLA and FIM circuit families on one IBM Heron backend and layout. The measured correlators deviate from exact references in a structured way, with the largest deviations in transverse edge channels. This constrains the mechanism interpretation of the earlier DLA/FIM hardware programme.

The result does not support quantum advantage, scalable tomography, full-state reconstruction, backend-general entanglement dynamics, or claims about unmeasured circuits, depths, layouts, or backends.

VII. REPRODUCIBILITY

The source code and data are in the `anulum/scpn-quantum-control` repository. The relevant commands are:

- live readiness and gated submission: `scripts/phase3_entanglement_tomography_ibm.py`;
- completed raw-count artefact: `data/phase3_entanglement_tomography/entanglement_tomography_live_ibm_marrakesh_2026-05-20T004334Z.json`;
- post-run reducer: `scripts/analyse_phase3_entanglement_tomography.py`;
- paper-asset generator: `scripts/generate_phase3_entanglement_paper_assets.py`.

VIII. CONCLUSION

A preregistered 166-circuit IBM Heron run measured 54 reduced-Pauli observables for promoted DLA and FIM Kuramoto–XY circuits. The resulting deviations from exact references are large and structured, with the strongest departures in transverse DLA edge correlators and a smaller but clear FIM $\lambda_{\text{fim}} = 4$ distortion. The paper's contribution is a bounded mechanism-separation result: it localises correlator-level hardware deformation accompanying the earlier leakage/retention observations, while avoiding scalable tomography, backend-general, or advantage claims.

[1] Y. Kuramoto, in *International Symposium on Mathematical Problems in Theoretical Physics* (Springer, 1975) pp. 420–422.

[2] J. A. Acebrón, L. L. Bonilla, C. J. P. Vicente, F. Ritort, and R. Spigler, Reviews of Modern Physics **77**, 137 (2005).